

Visualisation URL: <https://jabichebli.github.io/FIT3179/Visualisation2/index.html>

FDS URL:

<https://github.com/jabichebli/FIT3179/blob/main/Visualisation2/Five%20Design%20Sheets/5-Design-Sheets-Jason-Abi-Chebli.pdf>

Domain:

Earthquake Activity around Australia.

Why:

Users can explore historical earthquake patterns and trends, identify high-risk areas, and enhance awareness and preparedness for seismic events through visualisations of earthquake activity.

Who:

Following the recent earthquake swarm in the Hunter region, with over 30 quakes in September 2024, this visualisation serves Australians curious about seismic activity.

What:

1. Geoscience Australia (2024). Earthquakes@GA. [online] Ga.gov.au. Available at: <https://earthquakes.ga.gov.au/>.

The data was cleaned and normalised in Excel.

How:

Bar Chart: Presented at the top, it allows users to see the number of earthquakes over the years.

- *Marks:* Lines
- *Channels:* Position on Common Scale, Colour Hue

Proportional Symbol Map: This enables users to visualise where and how deep earthquakes have occurred. Custom features include filtering scroll bars to select the minimum and maximum magnitudes of interest.

- *Marks:* Points/Circles
- *Channels:* Area, Colour Hue, Colour Luminance

Donut Chart: This shows which states have historically experienced the greatest number of earthquakes.

- *Marks:* Area
- *Channels:* Tilt/Angle, Colour Hue

Choropleth Map: This displays the number of earthquakes per 100,000 km² in each state over the years, with customisation options for selecting different historical years.

- *Marks:* Geographic Regions (Area)
- *Channels:* Colour Hue, Colour Luminance

Population Pyramid: This allows users to compare the number of earthquakes per 100,000 km² between WA and VIC, illustrating that VIC experiences more per 100,000 km².

- *Marks:* Lines
- *Channels:* Position on Common Scale, Colour Hue

Bubble Plot: This compares the largest magnitude earthquake, the number of earthquakes, and the average earthquake magnitude for each state. Customisation options let users select the state, year, and minimum average magnitude of interest.

- *Marks:* Points
- *Channels:* Position on Common Scale, Area, Colour Hue